YouTube Shorts pack 1: five 30–60 second scripts (videos 2, 3, 6, 8, 10)

Vertical 9:16, text on screen, face or hands-on-whiteboard. Each script is about 120 words, roughly 45 seconds at a calm pace. **Shorts rules:** the hook in the first 2 seconds, one idea only, end with a loop or question. Title under 60 characters, plus #shorts and 2–3 tags.

Video 2: Speed vs velocity in 45 seconds

Title: Speed vs Velocity: the 45-second difference #shorts **On-screen hook:** "Same speed. Different velocity?!" **Script:** "Two cars, both at 60 km/h. Same speed, right? Yes. Same velocity? Not if one is going north and the other south! Speed is how fast. Velocity is how fast **and in which direction**. Here's the trick question. I run one full round of a 400 m track in 80 seconds. My average speed? 400 ÷ 80 = 5 m/s. My average velocity? I'm back where I started, so my displacement is zero. Velocity = zero! So, speed: distance ÷ time. Velocity: displacement ÷ time. Comment 'V' if you'd got that trick question wrong before 😄" **Text on screen:** Speed = distance/time · Velocity = displacement/time · Round trip → velocity 0 **Tags:** #class9science #motion #physics

Video 3: 3 diagram mistakes that cost marks

Title: 3 Diagram Mistakes That Cost You Marks #shorts **On-screen hook:** "Your diagram is right… but you still lost marks?" **Script:** "Three diagram mistakes I see in almost every answer sheet. One: label lines that cross each other. Examiners hate that. Keep them straight and parallel. Two: no arrows on rays in light diagrams. Without arrows, a ray diagram is just lines. Three: drawing in pen. Use a pencil so you can correct it neatly. Bonus: always write the **title** under the diagram, like 'Human heart: sectional view'. Fix these three and you'll pick up easy marks in every Science paper. Which diagram do you find hardest? Tell me below 👇" **Text on screen:** ❌ crossing labels · ❌ no arrows · ❌ pen · ✅ title **Tags:** #boardexam #class10science #studytips

Video 6: Atom vs molecule vs ion in 30 seconds

Title: Atom vs Molecule vs Ion in 30 Seconds #shorts **On-screen hook:** "3 words everyone mixes up" **Script:** "Atom: the smallest particle of an element, like one oxygen atom, O. Molecule: two or more atoms bonded together, like O_2 or H_2O. Ion: an atom that has gained or lost electrons, so it carries a charge, like Na^+ or Cl^-. Quick test: Is Na^+ an atom, a molecule or an ion? Ion! It lost one electron. Save this before your exam 📌" **Text on screen:** Atom = O · Molecule = O_2, H_2O · Ion = Na^+, Cl^- **Tags:** #class9science #chemistry #atoms

Video 8: One memory trick for cell organelles

Title: Cell Organelles: Never Forget Them Again #shorts **On-screen hook:** "Think of a cell as a factory 🏭" **Script:** "Here's how to remember cell organelles forever: the cell is a factory. The **nucleus** is the manager's office, where the instructions (DNA) are kept. **Mitochondria** are the power plant, making energy as ATP. **Ribosomes** are the workers who make proteins. The **ER** is

the corridors that carry materials around. The **Golgi apparatus** is the packing and dispatch department. **Lysosomes** are the cleaning staff: they can even destroy the factory if it's damaged. That's why they're called suicide bags! And the **cell membrane** is the security gate deciding who gets in and out. Which one did you forget most often? 👇" **Text on screen:** organelle → factory role, one at a time **Tags:** #class9science #biology #cell

Video 10: The question that tricks everyone

Title: 99% Get This Speed Question WRONG #shorts **On-screen hook:** "40 km/h there, 60 km/h back. Average speed?" **Script:** "I go to my friend's house at 40 km/h and come back at 60 km/h. Average speed? Most people say 50. It's NOT 50! Average speed = total distance ÷ total time. Say the distance is 120 km each way. Going takes 120 ÷ 40 = 3 hours. Coming back takes 120 ÷ 60 = 2 hours. Total: 240 km in 5 hours = **48 km/h**. You spend more time at the slower speed, so the average leans towards it. Did you say 50? Be honest in the comments 😄" **Text on screen:** ❌ 50 · ✅ 48 · total distance ÷ total time **Tags:** #maths #physics #class9science